

Velocity Loss as an Indicator of Neuromuscular Fatigue during Resistance Training

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ABSTRACT

SÁNCHEZ-MEDINA, L., and J. J. GONZÁLEZ-BADILLO. Velocity Loss as an Indicator of Neuromuscular Fatigue during Resistance Training. *Med. Sci. Sports Exerc.*, Vol. 43, No. 9, pp. 1725–1734, 2011. **Purpose:** This study aimed to analyze the acute mechanical and metabolic response to resistance exercise protocols (REP) differing in the number of repetitions (R) performed in each set (S) with respect to the maximum predicted number (P). **Methods:** Over 21 exercise sessions separated by 48–72 h, 18 strength-trained males (10 in bench press (BP) and 8 in squat (SQ)) performed 1) a progressive test for one-repetition maximum (1RM) and load–velocity profile determination, 2) tests of maximal number of repetitions to failure (12RM, 10RM, 8RM, 6RM, and 4RM), and 3) 15 REP (S × R[P]: 3 × 6[12], 3 × 8[12], 3 × 10[12], 3 × 12[12], 3 × 6[10], 3 × 8[10], 3 × 10[10], 3 × 4[8], 3 × 6[8], 3 × 8[8], 3 × 3[6], 3 × 4[6], 3 × 6[6], 3 × 2[4], 3 × 4[4]), with 5-min interset rests. Kinematic data were registered by a linear velocity transducer. Blood lactate and ammonia were measured before and after exercise. **Results:** Mean repetition velocity loss after three sets, loss of velocity pre-post exercise against the 1-m·s⁻¹ load, and countermovement jump height loss (SQ group) were significant for all REP and were highly correlated to each other ($r = 0.91–0.97$). Velocity loss was significantly greater for BP compared with SQ and strongly correlated to peak postexercise lactate ($r = 0.93–0.97$) for both SQ and BP. Unlike lactate, ammonia showed a curvilinear response to loss of velocity, only increasing above resting levels when R was at least two repetitions higher than 50% of P. **Conclusions:** Velocity loss and metabolic stress clearly differs when manipulating the number of repetitions actually performed in each training set. The high correlations found between mechanical (velocity and countermovement jump height losses) and metabolic (lactate, ammonia) measures of fatigue support the validity of using velocity loss to objectively quantify neuromuscular fatigue during resistance training. **Key Words:** MUSCLE STRENGTH, WEIGHT TRAINING, BLOOD LACTATE, AMMONIA, BENCH PRESS, FULL SQUAT

Knowledge of the mechanical and physiological aspects underlying resistance training (RT) is essential to improve our understanding of the stimuli that affect adaptation (8). Configuration of the exercise stimulus in RT has been traditionally associated with a combination of the so-called *acute resistance exercise variables* (exercise type and order, loading, number of repetitions and sets, rest duration, and movement velocity) (25,35). Although most of these variables have received considerable research attention, a question that remains ignored in the literature is the possibility of manipulating the number of repetitions actually performed in each set with respect to the maximum number that can be completed. It seems reasonable that the degree or level of effort is substantially different when performing, e.g., 8 of 12 possible repetitions with a given load (8[12]) compared with

performing all repetitions (12[12]). Lack of attention to this issue is likely due to an assumption that RT should always be performed to muscular failure. However, increasing evidence seems to suggest that reaching repetition failure may not necessarily improve the magnitude of strength gains (10,14,20,21). Furthermore, in the case of not exercising to failure, the optimal number of repetitions to perform under different loading conditions to achieve certain training goals has not been established.

Muscle fatigue is recognized as a complex, task-dependent and multifactorial phenomenon whose etiology is controversial and still a matter of much debate (12,13,29). Despite the many definitions of fatigue that have been proposed (2,4,12,13), a common element to most of them is the observation of an exercise-induced transient decline in muscle force-generating capacity. This decrease in force production is accompanied by an increase in the level of effort required to perform the exercise until eventually, if continued, task failure occurs (13,39). However, fatigue limits not only a fiber's capacity for maximal force generation but also the maximum velocity of shortening decreases and a slowing of relaxation occurs (2). Consequently, power output will be affected. In fact, an increased curvature of the force–velocity relationship is a major factor in the loss of muscle power (22). Therefore, all definitions of fatigue necessitate a decline in force, velocity, or power (39).

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During typical resistance exercise in isoinertial conditions, and assuming every repetition is performed with maximal voluntary effort, velocity unintentionally declines as fatigue develops (18). However, few studies analyzing the response to different RT schemes have described changes in repetition velocity or power (1,18,19,26). It thus seems necessary to conduct more research using models of fatigue that analyze the reduction in mechanical variables such as force, velocity, and power output over repeated dynamic contractions in actual training or competition settings (7,39).

Therefore, the purpose of the present study was to quantify the extent of neuromuscular fatigue while performing popular multijoint RT exercises for the upper (bench press) and lower body (squat) by analyzing the acute mechanical (velocity loss) and metabolic (blood lactate and ammonia) response to 15 types of resistance exercise protocols (REP) differing in the number of repetitions actually performed in each set with regard to the maximum predicted number. We hypothesized that both repetition velocity loss within a set and loss of velocity before versus immediately after exercise against a submaximal, individually determined, load would be highly correlated to indicators of metabolic stress and thus could be used to quantify the actual level of effort incurred during typical RT sessions.

METHODS

Subjects

Eighteen men (age = 25.6 ± 3.4 yr, body mass = 75.9 ± 9.1 kg, height = 176.6 ± 7.5 cm, body fat = $12.2\% \pm 3.7\%$) volunteered to take part in this study. Subjects were either professional firefighters or firefighter candidates with an RT experience ranging from 3 yr to beyond 5 yr. They were divided into two groups depending on the exercise to be performed: bench press (BP, $n = 10$) or full squat (SQ, $n = 8$). Initial one-repetition maximum (1RM) strength was 95.0 ± 14.9 kg for the BP and 97.1 ± 23.0 kg for the SQ group. In the 3 months preceding this study, subjects had been training two to three sessions per week and were capable of performing their respective exercise with proper technique. No physical limitations, health problems, or musculoskeletal injuries that could affect testing were found after a medical examination. None of the subjects were taking drugs, medications, or dietary supplements known to influence physical performance. The study was approved by the Research Ethics Committee of Pablo de Olavide University, and written informed consent was obtained from all subjects.

Study Design

During a period of approximately 8 wk, 21 exercise sessions were conducted in the following order: 1) an initial test with increasing loads for the individual determination of 1RM strength and full load-velocity relationship, 2) five tests of maximal number of repetitions to failure (XRM:

12RM, 10RM, 8RM, 6RM, 4RM), 3) 15 REP differing in the number of repetitions (R) actually performed in each set (S) with regard to the maximum predicted number of repetitions (P) ($S \times R[P]$: $3 \times 6[12]$, $3 \times 8[12]$, $3 \times 10[12]$, $3 \times 12[12]$, $3 \times 6[10]$, $3 \times 8[10]$, $3 \times 10[10]$, $3 \times 4[8]$, $3 \times 6[8]$, $3 \times 8[8]$, $3 \times 3[6]$, $3 \times 4[6]$, $3 \times 6[6]$, $3 \times 2[4]$, $3 \times 4[4]$). All these sessions were conducted on separate days, with 48 h of recovery time except the initial 1RM test, the XRM assessments, and the $3 \times 12[12]$, $3 \times 10[10]$, $3 \times 8[8]$, and $3 \times 6[6]$ REP (i.e., the most demanding protocols) after which 72 h of recovery was allowed. Sessions were performed in the evenings, at the same time of day for each participant and under similar environmental conditions (20°C – 22°C and 55%–65% humidity). During the present study, subjects did not perform any other RT besides some abdominal and lower-back strengthening exercises, and their endurance conditioning only consisted of running (BP group) or swimming (SQ group) twice per week (30 min at an intensity corresponding to 70%–80% of HR reserve).

Testing Procedures

Initial session and 1RM determination. An introductory session was used for body composition assessment, medical examination, and familiarization with testing protocols. Subjects arrived to the laboratory in the morning in a well-rested condition and fasted state. After being medically screened and their body composition determined, they carried out some practice sets with light and medium loads in their respective exercise (BP or SQ), while researchers emphasized proper technique. On the evening of the following day, individual load-velocity relationships and 1RM strength were determined using a progressive loading test. A detailed description of the BP testing protocol has been recently provided elsewhere (31). The BP was performed imposing a momentary pause (~ 1.5 s) at the chest between the eccentric and concentric actions to minimize the contribution of the rebound effect and allow for more reproducible, consistent measurements. In the SQ group, subjects started from the upright position with the knees and hips fully extended, stance approximately shoulder-width apart and the barbell resting across the back at the level of the acromion. Each subject descended in a continuous motion until the top of the thighs got below the horizontal (ground) plane, the posterior thighs and shanks making contact with each other, then immediately reversed motion and ascended back to the upright position. Auditory feedback based on eccentric distance traveled was provided to help each subject reach his previously determined squat depth. Unlike the eccentric phase that was performed at a normal, controlled speed, subjects were required to always execute the concentric phase of either BP or SQ in an explosive manner, at maximal intended velocity. Warm-up consisted of 5 min of stationary cycling at a self-selected easy pace, 5 min of static stretches and joint mobilization exercises, followed by two sets of eight and six repetitions

(3-min rest) with loads of 20 and 30 kg, respectively. Initial load was set at 20 kg for all subjects and was gradually increased in 10-kg increments until the attained mean propulsive velocity (MPV) was $<0.5 \text{ m}\cdot\text{s}^{-1}$ in the BP or $<0.8 \text{ m}\cdot\text{s}^{-1}$ in the SQ group. Thereafter, load was individually adjusted with smaller increments (5 down to 1 kg) so that 1RM could be precisely determined. The heaviest load that each subject could properly lift while completing full range of motion was considered to be his 1RM. Trained spotters were present when high loads were lifted to ensure safety. Three attempts were executed for light ($<50\%$ RM), two for medium (50% – 80% RM), and only one for the heaviest ($>80\%$ RM) loads. Interset rests ranged from 3 (light) to 6 min (heavy loads). Only the best repetition at each load, according to the criteria of fastest MPV (31), was considered for subsequent analysis.

Maximum repetition number assessment. For the XRM load assessments, subjects warmed up by performing four to five sets of five down to two repetitions (3-min rests), progressively increasing weight up to the load corresponding to $\sim 70\%$ (12RM), $\sim 75\%$ (10RM), $\sim 80\%$ (8RM), $\sim 85\%$ (6RM), or $\sim 90\%$ (4RM) of their previously determined 1RM. This was carefully controlled for each participant from his individual load–velocity profile because it has been recently shown that mean velocity can be used to precisely estimate loading intensity (15). After a 5-min rest, subjects completed one set to failure, whereas kinematic data from every repetition were registered.

Acute REP. The 15 types of REP were performed always using three sets and 5-min interset recoveries. Two measures were taken to ensure that the maximum predicted number of repetitions for each session was as accurate as possible. First, previous XRM assessments were used as a reference to individually determine absolute load for each REP. Second, because of the considerable number of exercise sessions undertaken in this study, strength levels were expected to change. Consequently, before starting each REP, adjustments in the proposed load (kg) were made when needed so that the velocity of the first repetition matched that expected from each subject's relative load–velocity relationship. In each session, subjects warmed up by performing three sets of six down to three repetitions (2-min rests) with increasing loads up to the individual load that elicited a $\sim 1.00 \text{ m}\cdot\text{s}^{-1}$ (1.04 ± 0.01 for SQ and 1.03 ± 0.01 for BP) MPV ($V_{1 \text{ m}\cdot\text{s}^{-1}}$). This value was chosen because it is a sufficiently high velocity, which is attained against medium loads ($\sim 45\%$ RM in BP and $\sim 60\%$ RM in SQ), and it allows a good expression of the effect of loading on velocity, besides being a relatively easy-to-move and well-tolerated load. The $V_{1 \text{ m}\cdot\text{s}^{-1}}$ load (kg) was thus taken as a preexercise reference measure against which to compare the velocity loss experienced after the three exercise sets. Subjects executed three maximal-effort consecutive repetitions against the $V_{1 \text{ m}\cdot\text{s}^{-1}}$ load right before starting the first set and again immediately after completing the last repetition of the third set (load was changed in 10–15 s with

the help of the spotters). Furthermore, the participants from the SQ group performed five maximal countermovement jumps (CMJ), separated by 20-s rests, right after executing the three preexercise repetitions with the $V_{1 \text{ m}\cdot\text{s}^{-1}}$ load and again after the final three postexercise repetitions with that load. On each occasion, CMJ height was registered, the highest and lowest values were discarded, and the resulting average was kept for analysis. Strong verbal encouragement and velocity feedback in every repetition was provided throughout all sessions to motivate participants to give a maximal effort.

Mechanical Measurements of Fatigue

Three different methods were used to quantify the extent of fatigue induced by each REP. The first method analyzed the decline in repetition velocity during the three consecutive exercise sets. It was calculated as the percent loss in MPV from the fastest (usually first) to the slowest (last) repetition of each set and averaged over the three sets. The second method examined the percent change in MPV pre–post exercise attained with the $V_{1 \text{ m}\cdot\text{s}^{-1}}$ load. The average MPV of the three repetitions before exercise was compared with the average MPV of the three repetitions after exercise, i.e., $100 (\text{average MPV}_{\text{post}} - \text{average MPV}_{\text{pre}}) / \text{average MPV}_{\text{pre}}$. Figure 1 shows an example of these velocity losses for a representative subject and protocol. The third method (only applied to the SQ group) involved the calculation of percent change in CMJ height pre–post exercise.

Blood Lactate and Ammonia Analyses

Capillary whole blood samples were drawn from the fingertip before exercise and immediately after each REP. Postexercise samples for the analysis of lactate ($5 \mu\text{L}$) were taken at 1, 3, and 5 min, whereas samples for ammonia ($20 \mu\text{L}$) were obtained at 1, 4, and 7 min during recovery to determine peak concentration. The Lactate Pro LT-1710 (Arkray, Kyoto, Japan) portable lactate analyzer was used

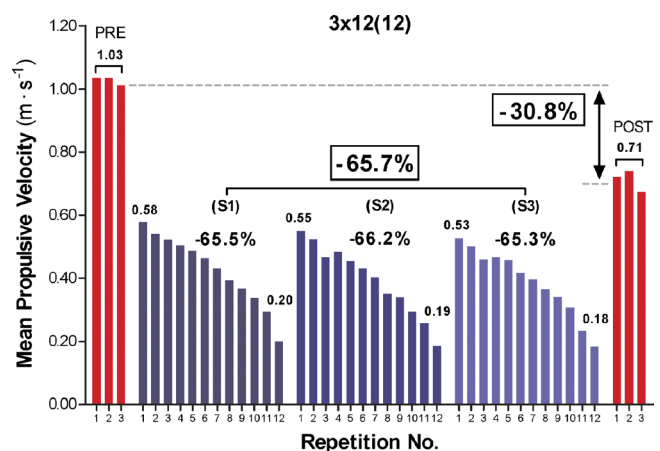


FIGURE 1—Example of quantification of percent velocity losses after a $3 \times 12[12]$ REP for a representative subject in the BP exercise. Both MPV loss over three sets (-65.7%) and MPV pre–post exercise against the $V_{1 \text{ m}\cdot\text{s}^{-1}}$ load (-30.8%) were calculated.

for lactate measurements. Ammonia was measured using PocketChem BA PA-4130 (Menarini Diagnostics, Florence, Italy). Both devices were calibrated before each exercise session according to the manufacturer's specifications. Reliability was calculated by assessing twice 15 different samples over the physiological range ($1.3\text{--}17.0\text{ mmol}\cdot\text{L}^{-1}$ for lactate and $35\text{--}150\text{ }\mu\text{mol}\cdot\text{L}^{-1}$ for ammonia). The coefficient of variation (CV) ranged from 2.6% to 4.1% for lactate and from 3.0% to 5.2% for ammonia.

Measurement Equipment and Data Acquisition

Height was measured to the nearest 0.5 cm during a maximal inhalation using a wall-mounted stadiometer (Seca 202; Seca Ltd., Hamburg, Germany). Body weight was determined, and fat percentage was estimated using an eight-contact electrode segmental body composition analyzer (Tanita BC-418; Tanita Corp., Tokyo, Japan). Jump height was measured using an infrared timing system (Optojump; Microgate, Bolzano, Italy). A Smith machine (Multipower Fitness Line, Peroga, Spain) that ensures a smooth vertical displacement of the bar along a fixed pathway was used for all sessions. A dynamic measurement system (T-Force System; Ergotech, Murcia, Spain) automatically calculated the relevant kinematic parameters of every repetition, provided auditory velocity and displacement feedback, and stored data on disk for analysis. This system consists of a linear velocity transducer interfaced to a personal computer using a 14-bit resolution analog-to-digital data acquisition board and custom software. Instantaneous velocity was sampled at a frequency of 1000 Hz and subsequently smoothed with a fourth-order low-pass Butterworth filter with a cutoff frequency of 10 Hz. A digital filter with no phase shift was then applied to the data. Validity and reliability were established by comparing the displacement measurements obtained by this device with a high-precision digital height gauge

(Mitutoyo HDS-H60C; Mitutoyo, Corp., Kawasaki, Japan) previously calibrated by the Spanish National Institute of Aerospace Technology. After performing the comparisons with 18 different T-Force units, the mean relative error in the velocity measurements was found to be $<0.25\%$, whereas displacement was accurate to $\pm 0.5\text{ mm}$. In addition, when simultaneously performing 30 repetitions with two devices (range = $0.3\text{--}2.3\text{ m}\cdot\text{s}^{-1}$ mean velocity), an intraclass correlation coefficient (ICC) of 1.00 (95% confidence interval = $1.00\text{--}1.00$) and CV of 0.57% were obtained for MPV, whereas an ICC of 1.00 (95% confidence interval = $0.99\text{--}1.00$) and CV of 1.75% were found for peak velocity. The velocities reported in the present study correspond to the mean velocity of the propulsive phase for each repetition. Mean propulsive values are preferable to mean concentric values because they avoid underestimating an individual's true neuromuscular potential when lifting light and medium loads, as well as being more stable and reliable than peak values (31). The propulsive phase was defined as that portion of the concentric phase during which barbell acceleration (a) is greater than acceleration due to gravity (i.e., $a > -9.81\text{ m}\cdot\text{s}^{-2}$).

Statistical Analysis

Correlations are reported using Pearson product-moment correlation coefficients (r). Relationships between mechanical losses and ammonia concentration were studied by fitting second-order polynomials to data. An independent-samples t -test was used to examine differences between exercises, whereas a related-samples t -test was used to analyze velocity and CMJ height pre-post changes as well as to compare preexercise and postexercise lactate and ammonia levels. Data are presented as mean $\pm$ SD. Significance was accepted at $P \leq 0.05$. Analyses were performed using SPSS software version 15.0 (SPSS, Chicago, IL).

TABLE 1. Mechanical and metabolic measurements of fatigue after each REP.

REP	Loss of MPV over Three Sets (%)		Loss of MPV with $V_{1\text{ m}\cdot\text{s}^{-1}}$ Load (%)		Loss of CMJ Height (%)	Lactate ($\text{mmol}\cdot\text{L}^{-1}$)		Ammonia ($\mu\text{mol}\cdot\text{L}^{-1}$)	
	SQ	BP	SQ	BP		SQ	BP	SQ	BP
3 $\times$ 12[12]	46.5 $\pm$ 3.8	***63.3 $\pm$ 4.0	21.3 $\pm$ 9.1	**32.8 $\pm$ 8.5	19.3 $\pm$ 4.4	12.5 $\pm$ 1.9	***8.2 $\pm$ 1.3	†††125 $\pm$ 34	†††111 $\pm$ 20
3 $\times$ 10[12]	37.1 $\pm$ 7.7	***51.1 $\pm$ 5.5	14.6 $\pm$ 5.5	**24.9 $\pm$ 6.9	16.6 $\pm$ 3.9	10.6 $\pm$ 1.2	***6.7 $\pm$ 1.0	†62 $\pm$ 14	†††71 $\pm$ 11
3 $\times$ 8[12]	32.3 $\pm$ 7.6	36.5 $\pm$ 4.3	10.6 $\pm$ 1.9	**15.3 $\pm$ 3.6	11.4 $\pm$ 1.9	8.0 $\pm$ 1.4	**5.7 $\pm$ 1.4	49 $\pm$ 16	49 $\pm$ 18
3 $\times$ 6[12]	20.2 $\pm$ 4.3	*24.2 $\pm$ 2.3	9.7 $\pm$ 2.1	8.1 $\pm$ 1.4	9.6 $\pm$ 1.4	4.9 $\pm$ 0.3	4.2 $\pm$ 0.9	46 $\pm$ 14	45 $\pm$ 12
3 $\times$ 10[10]	45.7 $\pm$ 7.0	***58.4 $\pm$ 4.5	21.0 $\pm$ 8.9	*30.5 $\pm$ 8.3	17.0 $\pm$ 3.6	11.7 $\pm$ 2.2	***7.8 $\pm$ 1.2	†††97 $\pm$ 19	†††89 $\pm$ 16
3 $\times$ 8[10]	32.3 $\pm$ 5.5	***46.1 $\pm$ 4.2	15.2 $\pm$ 4.3	17.7 $\pm$ 3.9	13.6 $\pm$ 1.9	8.6 $\pm$ 1.3	***6.0 $\pm$ 0.6	62 $\pm$ 20	††64 $\pm$ 17
3 $\times$ 6[10]	22.0 $\pm$ 8.0	*29.8 $\pm$ 4.5	11.0 $\pm$ 4.1	13.6 $\pm$ 2.6	10.3 $\pm$ 2.1	6.3 $\pm$ 1.6	**4.6 $\pm$ 0.8	48 $\pm$ 10	47 $\pm$ 13
3 $\times$ 8[8]	39.8 $\pm$ 4.0	***56.9 $\pm$ 3.7	18.2 $\pm$ 5.0	*26.8 $\pm$ 7.9	15.8 $\pm$ 4.0	10.4 $\pm$ 2.1	**7.5 $\pm$ 1.4	†78 $\pm$ 28	†††79 $\pm$ 20
3 $\times$ 6[8]	29.4 $\pm$ 9.4	*39.0 $\pm$ 4.5	10.5 $\pm$ 1.2	*13.9 $\pm$ 3.2	11.3 $\pm$ 2.2	7.1 $\pm$ 2.1	**4.8 $\pm$ 1.0	52 $\pm$ 22	54 $\pm$ 19
3 $\times$ 4[8]	21.2 $\pm$ 8.6	24.8 $\pm$ 2.9	9.2 $\pm$ 1.7	9.6 $\pm$ 1.5	6.0 $\pm$ 2.0	4.5 $\pm$ 0.8	*3.4 $\pm$ 0.9	42 $\pm$ 11	49 $\pm$ 19
3 $\times$ 6[6]	41.9 $\pm$ 4.9	***56.8 $\pm$ 5.7	14.7 $\pm$ 5.4	**24.7 $\pm$ 6.6	13.2 $\pm$ 2.6	10.0 $\pm$ 1.7	***6.9 $\pm$ 1.1	65 $\pm$ 23	††68 $\pm$ 14
3 $\times$ 4[6]	28.1 $\pm$ 6.1	*33.8 $\pm$ 3.6	10.3 $\pm$ 3.0	9.2 $\pm$ 2.8	9.9 $\pm$ 2.7	5.2 $\pm$ 0.9	*4.0 $\pm$ 0.9	56 $\pm$ 21	52 $\pm$ 17
3 $\times$ 3[6]	19.6 $\pm$ 7.1	23.7 $\pm$ 3.0	8.0 $\pm$ 2.6	7.2 $\pm$ 1.4	6.4 $\pm$ 2.2	3.5 $\pm$ 0.6	3.1 $\pm$ 0.7	47 $\pm$ 16	51 $\pm$ 21
3 $\times$ 4[4]	32.0 $\pm$ 5.1	***49.8 $\pm$ 6.6	9.5 $\pm$ 1.9	***18.8 $\pm$ 3.8	10.6 $\pm$ 3.4	6.9 $\pm$ 1.7	**4.9 $\pm$ 1.0	61 $\pm$ 28	53 $\pm$ 16
3 $\times$ 2[4]	16.6 $\pm$ 4.5	18.9 $\pm$ 4.4	6.1 $\pm$ 1.8	5.2 $\pm$ 1.2	5.7 $\pm$ 1.3	3.0 $\pm$ 0.6	2.6 $\pm$ 0.7	41 $\pm$ 12	48 $\pm$ 14

Data are mean $\pm$ SD. Losses are reported as positive values.

BP, bench press ($n = 10$); SQ, squat ($n = 8$).

Significant differences between SQ and BP exercises: * $P \leq 0.05$, ** $P \leq 0.01$, *** $P \leq 0.001$.

Significant differences between preexercise and postexercise ammonia: † $P \leq 0.05$, †† $P \leq 0.01$, ††† $P \leq 0.001$.

Postexercise lactate significantly different ($P \leq 0.001$) from preexercise for all REP and both exercises except $3 \times 2[4]$ in BP ($P \leq 0.01$).

Loss of MPV over three sets, loss of MPV with $V_{1\text{ m}\cdot\text{s}^{-1}}$ load, and loss of CMJ height were significant ($P \leq 0.001$) for all REP and both exercises.

RESULTS

Velocity and CMJ height losses. Both percent loss of MPV over three sets and loss of MPV pre-post exercise with the $V_{1 \text{ ms}^{-1}}$ load, gradually increased as the number of performed repetitions in each set approached the maximum predicted number of repetitions for each type of REP (Table 1). **Velocity losses were significantly greater for BP compared with SQ for most types of REP.** This difference in the magnitude of loss of velocity between exercises increased as the number of performed repetitions approached the maximum (Table 1). MPV losses, both over three sets and pre-post with $V_{1 \text{ ms}^{-1}}$ load, were statistically significant ($P \leq 0.001$) for all REP and both exercises. The decrease in CMJ height pre-post exercise was greater as the number of performed repetitions approached the maximum for each REP. Postexercise CMJ height was significantly different ($P \leq 0.001$) from preexercise after all REP.

Relationships between mechanical measurements of fatigue. A very high correlation was found between relative loss of MPV over three sets and loss of MPV pre-post exercise with the $V_{1 \text{ ms}^{-1}}$ load for both SQ ($r = 0.91$; Fig. 2A) and BP ($r = 0.97$; Fig. 2B) exercises. For the SQ group, similarly high correlations were found between percent loss of CMJ height pre-post exercise and (a) MPV loss over three sets ($r = 0.92$, $P \leq 0.001$; Fig. 3A) and (b) loss of MPV pre-post with the $V_{1 \text{ ms}^{-1}}$ load ($r = 0.93$, $P \leq 0.001$; Fig. 3B).

Blood lactate and ammonia response. Peak postexercise lactate concentration linearly increased as the number of performed repetitions in each set approached the maximum predicted number of repetitions, both in SQ and in BP (Table 1). For any REP, lactate levels were always higher after the SQ compared with the BP exercise, these differences being significant for most of the protocols analyzed (Table 1). Postexercise ammonia levels were significantly higher than preexercise resting values for the 3×12 [12], 3×10 [12], 3×10 [10], 3×8 [10], 3×8 [8], and 3×6 [6] REP in BP; and 3×12 [12], 3×10 [12], 3×10 [10], and 3×8 [8] REP in SQ (Table 1). Peak postexercise ammonia concentration did not increase above basal resting values ($\leq 50 \mu\text{mol}\cdot\text{L}^{-1}$) when the number of performed repetitions in each set was half the maximum predicted number. No statistically significant differences in postexercise ammonia were found between SQ and BP for any REP.

Relationships between mechanical and metabolic measures of fatigue. A nearly perfect correlation between MPV loss over three sets and postexercise peak lactate was found for both SQ ($r = 0.97$, $P < 0.001$) and BP ($r = 0.95$, $P < 0.001$) exercises (Fig. 4A). Very high correlations were also found between loss of MPV pre-post exercise with the $V_{1 \text{ ms}^{-1}}$ load and lactate for SQ ($r = 0.93$, $P < 0.001$) and BP ($r = 0.97$, $P < 0.001$) (Fig. 4C). Unlike lactate, which linearly increased with greater velocity loss (Figs. 4A, C), the response of ammonia to loss of velocity followed a curvilinear relationship and better fitted a quadratic regression (Figs. 4B, D). Thus, from a MPV loss

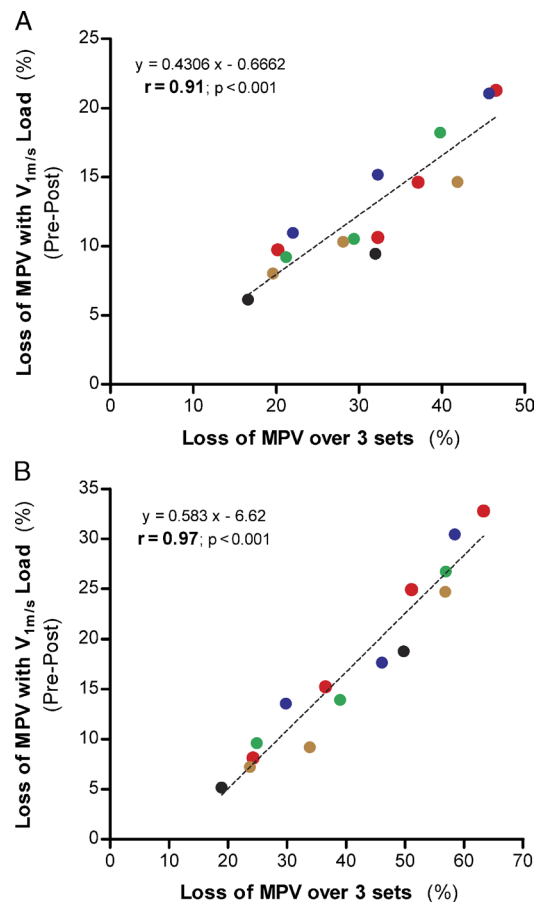


FIGURE 2—Relationships between relative loss of MPV over three sets and loss of MPV pre-post exercise against the $V_{1 \text{ ms}^{-1}}$ load in SQ (A) and BP (B) exercises. Each data point corresponds to one of the 15 different REP analyzed. Different symbol colors are used to differentiate between the maximum predicted number of repetitions (P) for each REP: black (P = 4), brown (P = 6), green (P = 8), blue (P = 10), and red (P = 12).

over three sets of $\sim 30\%$ (SQ) or $\sim 35\%$ (BP), blood ammonia levels started to increase steadily above resting values (Fig. 4B). When considering the loss of MPV pre-post with the $V_{1 \text{ ms}^{-1}}$ load, the magnitudes of velocity loss from which ammonia increased above resting values were $\sim 15\%$ (SQ) and $\sim 20\%$ (BP) (Fig. 4D). Percent loss of CMJ height pre-post exercise was highly correlated with lactate ($r = 0.97$, $P < 0.001$; Fig. 3C). Ammonia showed a curvilinear response to loss of CMJ height so that from $\sim 12\%$ loss of CMJ height, ammonia increased steadily above resting levels (Fig. 3D).

DISCUSSION

To the best of our knowledge, this is the first study to analyze the acute response to manipulating the number of repetitions actually performed in each training set with regard to the maximum number of repetitions that can be completed. Although some research has compared the effect of failure versus nonfailure training approaches on strength gains (9,10,14,20,21,38), the mechanical and metabolic

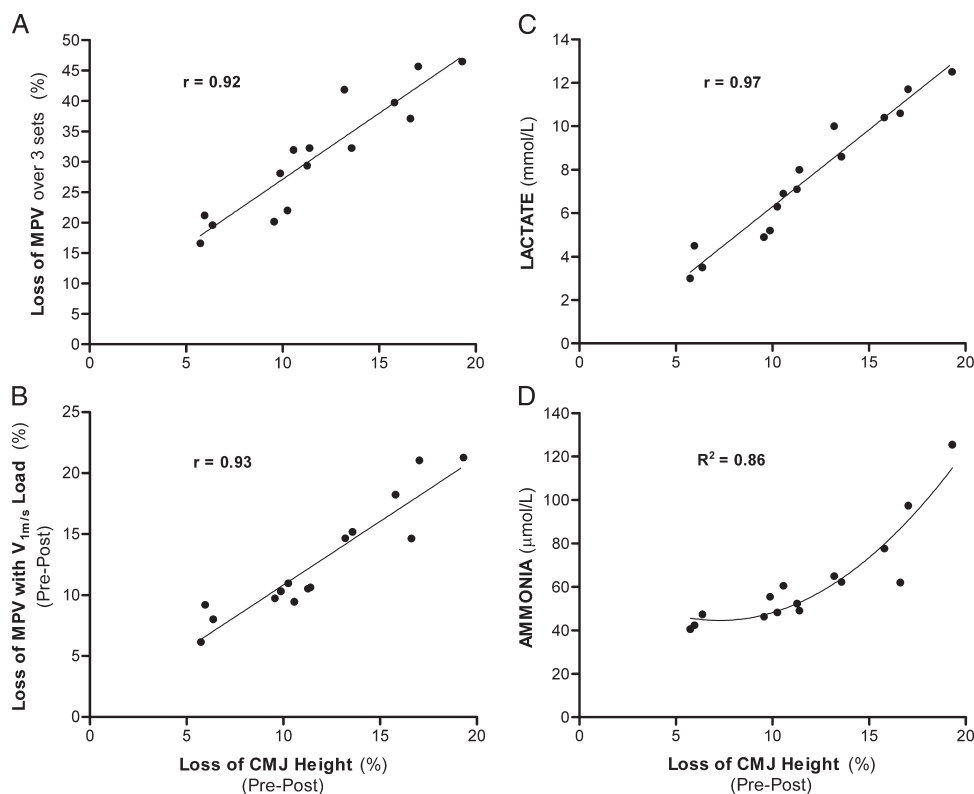


FIGURE 3—Relationships between relative loss of CMJ height pre-post exercise and loss of MPV over three sets (A), loss of MPV pre-post exercise against the $V_{1m/s}$ load (B), lactate (C), and ammonia (D) for the SQ exercise group. Each data point corresponds to one of the 15 different REP analyzed.

responses to different repetition schemes in which a set is ended before reaching muscular failure had not been previously analyzed. In the present study, a detailed examination of 15 different types of REP was conducted under controlled conditions to assess whether loss of repetition velocity could be used as an objective indicator of the extent of neuromuscular fatigue induced by typical RT sessions. Our results indicate that, by monitoring repetition velocity during training, it is possible to reasonably estimate the metabolic stress and neuromuscular fatigue induced by resistance exercise. A unique finding of this study is that ammonia, unlike lactate, shows a curvilinear response to loss of repetition velocity during RT. Some REP, especially those consisting of eight or more repetitions per set leading to failure (3×12 [12], 3×10 [10], and 3×8 [8]), caused ammonia to significantly rise above resting values, which could indicate an accelerated purine nucleotide degradation, thereby suggesting that such protocols may require longer recovery times.

Most of the literature examining neuromuscular fatigue has traditionally used isolated muscle preparations, both *in vitro* and *in situ*, as well as electrically stimulated muscle fibers. Isometric or isokinetic contractions made before and immediately after the fatiguing task, as well as during the activity, have been commonly used to quantify fatigue (27,29). Although such laboratory experiments are certainly necessary to identify the physiological mechanisms under-

lying the onset of muscle fatigue, they bear little resemblance to the majority of muscle actions performed in actual sports training and competition settings. Hence, there is a need to use fatigue protocols and outcome measures closer to isoinertial *in vivo* training movements (7,27). Because fatigue is postulated to be a continuous rather than a failure-point phenomenon (7), the gradual decrease in repetition velocity that takes place during repeated dynamic contractions can be interpreted as evidence of impaired neuromuscular function and its measurement could provide a relatively simple yet objective means of quantifying the extent of fatigue.

The present study confirms that the magnitude of velocity loss experienced during RT gradually increases as the number of performed repetitions in a set approaches the maximum predicted number. This was an expected result because it is known that velocity naturally slows down during a training set as fatigue develops (11,18,26). However, to the authors' knowledge, the actual values of velocity loss (Table 1) after a wide range of REP performed within the most typical RT intensity range ($\sim 70\%$ – 90% 1RM) had not been previously described. A finding worth noting is that greater MPV losses were experienced for BP compared with SQ for all protocols analyzed (Table 1). This is in agreement with previous results from Izquierdo et al. (18) who compared the pattern of repetition velocity decline when performing sets to failure with loads corresponding to 60%,

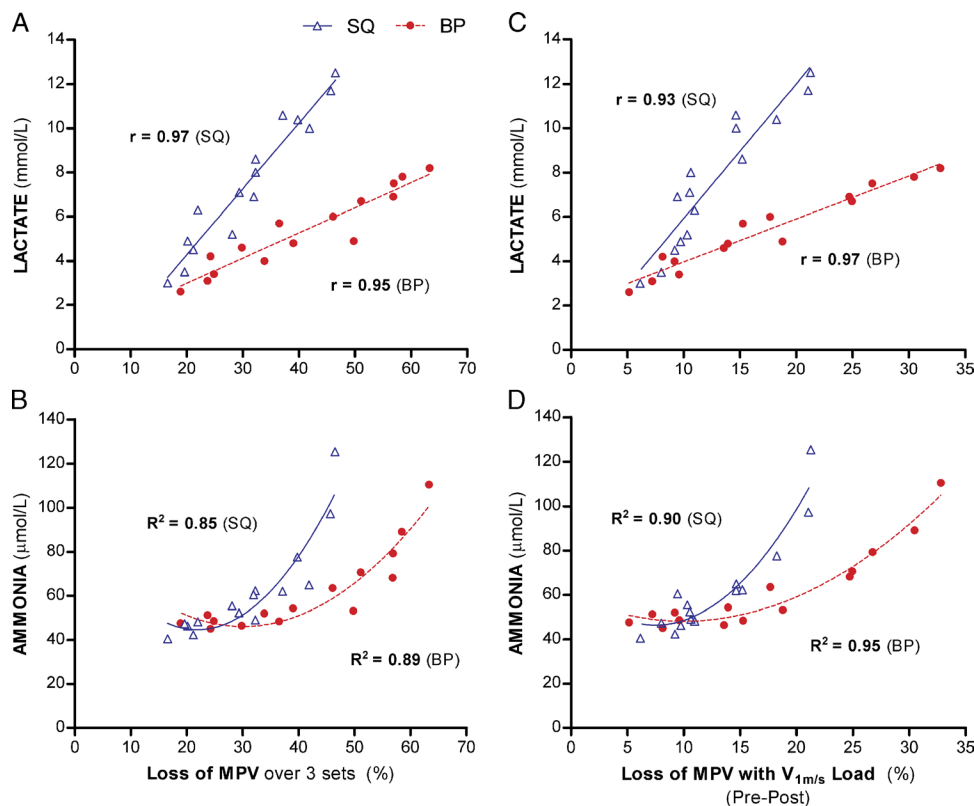


FIGURE 4—Relationships between relative loss of MPV over three sets and peak postexercise: lactate (A) and ammonia (B); and between MPV pre-post exercise against the $V_{1\text{ m/s}}$ load and lactate (C) and ammonia (D) for the BP and SQ exercises. Each data point corresponds to one of the 15 different REP analyzed.

65%, 70%, and 75% 1RM in the BP and half-squat exercises. The greater velocity loss in the BP could be because 1RM for this exercise is attained at a considerably slower mean velocity ($\sim 0.16\text{ m}\cdot\text{s}^{-1}$) than that for the SQ ($\sim 0.35\text{ m}\cdot\text{s}^{-1}$) (15,18). The lower 1RM mean velocity in the BP could be related to the greater movement control and smaller muscle groups involved in this exercise (more localized fatigue) compared with the SQ (fatigue distributed among a greater amount of muscle mass). The relative position of the “sticking region” in these exercises may also explain these velocity differences, as the squat allows more time/distance for force production after such region. We must finally consider that the BP was performed in a concentric-only (no rebound) action, whereas the SQ exercise is influenced by the stretch-shortening cycle that takes place when transitioning from an eccentric to a concentric action.

In essence, all models of fatigue entail two components: fatigue induction and fatigue quantification (27). In the present study, fatigue was quantified using two different methods: 1) percent decline in MPV over the three consecutive exercise sets and 2) percent change in MPV pre-post exercise attained with the $V_{1\text{ m/s}}$ load, as well as percent change in CMJ height pre-post (SQ group only). Because fatigue has been traditionally defined as a loss of force-generating capability with an eventual inability to sustain exercise at the required or expected level (4,13), the post-

exercise decline in movement velocity experienced against a given submaximal load (in this case, the $V_{1\text{ m/s}}$ load) can be considered as a good expression of neuromuscular fatigue. Indeed, in addition to force reduction, other aspects of neuromuscular performance that are affected by fatigue are muscle-shortening velocity (decreases) and relaxation time (increases) (2). Because of fatigue, the load that was lifted at $\sim 1.00\text{ m}\cdot\text{s}^{-1}$ in a rested, preexercise state, will be moved at a considerably slower velocity after the REP. The subject will undoubtedly perceive a greater effort when moving the same absolute load in the fatigued state, a situation that corresponds well with the definition of Enoka and Stuart (13). Besides being a relatively easy-to-move and well-tolerated load for most RT exercises, the $V_{1\text{ m/s}}$ load is quick to determine as part of the warm-up and facilitates the calculation of percentage losses.

Similar to loss of MPV over three sets, the magnitude of loss of MPV pre-post with the $V_{1\text{ m/s}}$ load gradually increased as the number of performed repetitions in each set approached the maximum predicted number for each type of REP (Table 1). Relative loss of velocity with the $V_{1\text{ m/s}}$ load was of lesser magnitude than MPV loss over three sets and higher for BP compared with SQ, especially as the number of performed repetitions increased toward maximum (Table 1). The same pattern of decline was observed when analyzing loss of CMJ height pre-post exercise for

the SQ group, which seems to follow the same rationale. Loss of CMJ height is equivalent to loss of vertical velocity at take-off, so in essence we are quantifying fatigue by the loss of muscle-shortening velocity. Several studies have used measurements of vertical jump height pre-post exercise to quantify the extent of fatigue. Smilios (33) observed CMJ height losses of 33% and 23% after exercise to failure in the leg press with loads of 70% and 90% RM, respectively. These reductions are greater than those obtained in the present study (19% and 11%) in the equivalent REP of 12[12] (~70% RM) and 4[4] (~90% RM). However, in Smilios's study (33), participants were not required to perform each repetition with maximal voluntary effort, and the number of repetitions actually performed with each load was not reported, which makes it difficult to compare with our data. Rodacki et al. (30) induced fatigue by requesting subjects to extend and flex their knees to failure in a weight machine. The loads used corresponded to ~50% (extensors) and ~40% (flexors) of each subject's body mass. Mean losses in CMJ height of 14% (extensors) and 6% (flexors) were found. Data from Rodacki et al. (30) suggest that the incurred fatigue and degree of effort was highly variable between participants (~10–26 repetitions in extension; ~18–36 repetitions in flexion) and thus precludes direct comparison with our data. Gorostiaga et al. (16) examined CMJ height loss after typical sprint training workouts in 400-m elite runners. They found reductions of 5%–19% in CMJ height pre-post exercise, with no clear relationship to sprint distance. Comparing our findings with those of these investigations is difficult because the protocols used to induce fatigue, the samples, and even the type of actions and movement velocities greatly differed between studies. Nevertheless, it seems clear from this body of research that loss of CMJ height can be used as an indicator of neuromuscular fatigue.

In the present study, very high and significant correlations ($r = 0.91$ – 0.97) were found between the three different types of mechanical measures used to assess neuromuscular fatigue (Figs. 2 and 3A, B). These relationships are an important finding for the quantification and monitoring of training load during RT. The fact that there exists such a close relationship between loss of MPV over three sets and loss of MPV with the $V_{1 \text{ ms}^{-1}}$ load in two exercises as different as SQ (Fig. 2A) and BP (Fig. 2B), as well as between both variables and loss of CMJ height in the SQ group (Figs. 3A, B), is a novel finding that emphasizes the validity of using percent loss of repetition velocity within a set as an indicator of neuromuscular fatigue. The relationships observed in Figure 2 also mean that, for a given percent loss of velocity within a set, the degree of fatigue incurred during RT is very similar irrespective of the number of repetitions the subject is able to perform (shown in different colors in Fig. 2), at least in a range from 4 (~90% RM) to 12 (~70% RM) repetitions.

The validity of using percent velocity loss to quantify neuromuscular fatigue during RT is further supported by the relationships observed between mechanical measures

of fatigue and metabolic stress (acute lactate and ammonia responses) (Figs. 3C, D and 4). Lactate increased linearly as the number of performed repetitions approached the maximum predicted for each type of REP (Table 1) and showed extremely high correlations ($r = 0.93$ – 0.97) with loss of MPV over three sets (Fig. 4A), loss of MPV pre-post exercise with the $V_{1 \text{ ms}^{-1}}$ load (Fig. 4C), and loss of CMJ height (Fig. 3C). The highest peak lactate values (~10.5–12.5 mmol·L⁻¹ in SQ and ~7.5–8.0 mmol·L⁻¹ in BP) were obtained when performing 8–12 repetitions per set. Lactate levels were significantly higher for SQ than BP after most REP analyzed (Table 1), which can be attributed to the greater muscle mass involved in the full squat. The REP that resulted in the highest lactate response were 3×12 [12], 3×10 [12], 3×10 [10], and 3×8 [8], i.e., the type of protocols commonly used to induce muscle hypertrophy, which is in line with previous research (23,24,34). Interestingly, peak lactate values after the 3×6 [12] BP protocol (4.2 ± 0.9 mmol·L⁻¹) were very similar to those found by Abdessemed et al. (1) when performing 10×6 [12] under different interset recovery conditions. They found that blood lactate did not significantly increase after the third set when using 3-min (4.7 ± 0.8 mmol·L⁻¹) or 5-min (3.6 ± 0.7 mmol·L⁻¹) rests. However, the 1-min rest condition resulted in a significantly greater lactate elevation in sets 4 to 10, concomitant with much greater reductions in mean repetition power output.

A unique and interesting finding of the present study is that ammonia response, unlike lactate, shows a curvilinear relationship to loss of velocity (Figs. 4B, D) and seems independent of the exercise (BP or SQ). Peak postexercise ammonia only increased above basal resting levels when the number of performed repetitions in each set was at least two higher than half the maximum predicted number (Table 1), thus suggesting the existence of a certain “level of effort threshold” to be exceeded for blood ammonia to respond. This nonlinear response of ammonia is similar to that found in some studies, which analyzed the physiological response to incremental exercise (3,6,32), but to our knowledge, it had not been previously documented for RT. An increase in blood ammonia levels during short-term high-intensity exercise is usually interpreted as indicative of an accelerated ammonia production by muscle resulting from the deamination of AMP to IMP. A loss of purines has been documented after high-intensity exercise sessions such as repeated sprints (17,36). Because *de novo* synthesis of nucleotides is a slow and energy-consuming process, muscle performance can remain significantly reduced up to 48–72 h after exercise (17). According to the results of this study, the REP that resulted in blood ammonia significantly higher than resting levels were 3×12 [12], 3×10 [12], 3×10 [10], and 3×8 [8] in SQ and 3×12 [12], 3×10 [12], 3×10 [10], 3×8 [10], 3×8 [8], and 3×6 [6] in BP, with considerably greater values for those leading to failure in each set (Table 1). It seems plausible to suggest that these types of protocols may cause an accelerated purine nucleotide

degradation that would increase the amount of time needed for recovery after training. The mean peak postexercise ammonia concentrations in this study ($125 \mu\text{mol}\cdot\text{L}^{-1}$ in SQ, $110 \mu\text{mol}\cdot\text{L}^{-1}$ in BP for the $3 \times 12[12]$ REP) are similar to those obtained by Izquierdo et al. (19) but lower than the extremely high values ($>200 \mu\text{mol}\cdot\text{L}^{-1}$) found after $3 \times 10\text{RM}$ or $3 \times 5\text{RM}$ in multiple exercises with only 1-min recovery (24).

Although some studies have reported the point within a set where a significant reduction in velocity (18) or power output (1,26) was observed, the optimal time to terminate a set before reaching failure has never been clearly established. Although the present study does not come up with a definitive answer to that question, it does, however, provide us with some valuable information that may indicate when it could be appropriate to end a set. According to our results (Table 1; Figs. 3 and 4), a maximum MPV loss of $\sim 30\%$ for SQ and $\sim 35\%$ for BP could be established to prevent blood ammonia to significantly rise above resting levels. These theoretical thresholds for velocity loss could be used as a preliminary reference to undertake a longitudinal study aimed to examine the effect of training with different repetition velocity losses (e.g., 15%, 30%, and 45%) on neuromuscular performance (1RM strength, rate of force development, maximal power production, etc.).

Monitoring repetition velocity during resistance exercise seems important because both the neuromuscular demands and the training effect itself largely depend on the velocity at which loads are lifted. A velocity- or power-based approach to RT is not entirely new, and authors such as Bosco (5) and Tidow (37) already provided some initial guidelines for putting it into practice. However, the role placed by movement velocity has not been sufficiently investigated (28). The findings obtained in the present study strongly support the use of velocity monitoring to control the degree of incurred fatigue. Because loads must be specific to ensure an optimal training stimulus, setting a certain velocity loss threshold during RT can serve to avoid performing unnecessary repetitions that may not be contributing to the desired training effect. Furthermore, the immediate velocity feed-

back the athlete receives during each session may increase the potential for adaptation. With this training approach, instead of a certain amount of weight to be lifted, strength and conditioning coaches should prescribe resistance exercise in terms of two variables: 1) first repetition's mean velocity, which is intrinsically related to loading intensity (15); and 2) a maximum percent velocity loss to be allowed in each set. When this percent loss limit is exceeded the set must be terminated. The limit of repetition velocity loss should be set beforehand depending on the primary training goal being pursued, the particular exercise to be performed, as well as the training experience and performance level of the athlete. More studies are warranted to further explore this velocity-based approach to RT.

In conclusion, the present data show that the relationship between the number of repetitions actually performed in a set and the maximum predicted number that can be completed is an important aspect to take into account when prescribing resistance exercise because the velocity loss and metabolic stress clearly differ when manipulating these variables. The high correlations found between mechanical (velocity and CMJ height losses) and metabolic (lactate, ammonia) measures of fatigue support the validity of using velocity loss to objectively quantify neuromuscular fatigue during RT. The nonlinear response of blood ammonia to loss of repetition velocity could perhaps be used as a reference to indicate the point within a set where the exercise should be terminated when the main training objective is to improve movement velocity or maximal power production. Future experimental research should compare the effects of training with different magnitudes of velocity loss on neuromuscular performance. The present study is expected to contribute to the field of exercise science by allowing a more rational characterization of the RT stimulus.

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